

CHAPTER THREE

HYPOTHESIS TESTING

3.1. Basic Concepts

What is a Hypothesis?

A hypothesis is a statement about a population. Data are then used to check the reasonableness of the statement. In statistical analysis we make a claim, that is, state a hypothesis, collect data, and then use the data to test the assertion.

Hypothesis is a statement about a population parameter developed for the purpose of testing.

In most cases the population is so large that it is not feasible to study all the items, objects, or persons in the population. As a result, an alternative to measuring or interviewing the entire population is to take a sample from the population. We can, therefore, test a statement to determine whether the sample does or does not support the statement concerning the population.

What is Hypothesis Testing?

Hypothesis testing is a procedure based on sample evidence and probability theory to determine whether the hypothesis is a reasonable statement. This is also one way of making inference about population parameter, where the investigator has prior notion about the value of the parameter.

Hypothesis testing starts with a statement, or assumption, about a population parameter such as the population mean. The major purpose of hypothesis testing is to choose between two competing hypotheses about the value of a population parameter.

3.2. Steps in Testing Hypothesis

There is a five-step procedure that systematizes hypothesis testing.

- Step 1 - State null and alternate hypotheses
- Step 2 - Select a level of significance
- Step 3 - Identify the test statistic
- Step 4 - Formulate a decision rule
- Step 5 - Make a decision

Step 1: State the Null Hypothesis (H_0) and the Alternate Hypothesis (H_1)

The first step is to state the hypothesis being tested. It is called the null hypothesis, designated H_0 .

Null Hypothesis

- ✓ Generally speaking, the null hypothesis is developed for the purpose of testing.
- ✓ The null hypothesis is a statement that is not rejected unless our sample data provide convincing evidence that it is false.

Null Hypothesis is a statement about the value of a population parameter.

- ✓ It is the hypothesis of equality or the hypothesis of no difference.

Alternate Hypothesis

- ✓ Alternate hypothesis is a statement that is accepted if the sample data provide sufficient evidence that the null hypothesis is false.
- ✓ It is assumed to be true when the null hypothesis is false.
- ✓ It is the hypothesis of difference.
- ✓ Usually denoted by H_1 or H_a .

Both the null and alternative hypothesis should be stated before any statistical test of significance is conducted.

Example: A recent article indicated that the mean age of U.S. commercial aircraft is 15 years. To conduct a statistical test regarding this statement:

Step 1: Determine the null and alternative hypotheses

The H_0 represents the current or reported condition.

$$H_0: \mu = 15$$

The alternative hypothesis is that the statement is not true. That is,

$$H_1: \mu \neq 15$$

N.B. The null hypothesis will always contain the equal sign, but which will never appear in the alternative hypothesis.

Step 2: Select a Level of Significance

Level of significance is the probability of rejecting the null hypothesis when it is true.

The level of significance is designated α , the Greek letter alpha. It is also sometimes

called the level of risk. This means that it is the risk you take of rejecting the null hypothesis when it is really true. The researcher must decide on the level of significance before formulating a decision rule and collecting sample data.

- ✓ By rejecting a true null hypothesis, we committed a Type I error. The probability of committing a Type I error is α .

Type I Error - Rejecting the null hypothesis, H_0 , when it is true.

$$\alpha = \text{Probability of Type I error} = P(\text{rejecting } H_0 \mid H_0 \text{ is true})$$

- ✓ The probability of committing another type of error, called a Type II error, is designated by the Greek letter beta (β).

Type II Error - Accepting the null hypothesis when it is false.

$$\beta = \text{Probability of Type II error} = P(\text{accepting } H_0 \mid H_0 \text{ is false})$$

- ✓ The following table summarizes the decisions the researcher could make and the possible consequences.

	<u>Researcher</u>	
Null Hypothesis	Accept H_0	Reject H_0
H_0 is true	Correct decision	Type I error
H_0 is false	Type II error	Correct decision

Step 3: Select the Test Statistic

- ✓ There are many test statistics. In this chapter we use both z and t as the test statistic.
- ✓ Tests statistic is a value, determined from sample information, used to determine whether to reject the null hypothesis.
- ✓ In hypothesis testing for the mean (μ) when σ is known or the sample size is large, the test statistic z is computed by: $z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$
- ✓ The z value is based on the sampling distribution of $\bar{X}$, which follows the normal distribution when the sample is reasonably large.
- ✓ t test is used for small sample case ($n < 30$) and population standard deviation is not known (i.e., sample standard error is known).

Step 4: Formulate the Decision Rule

A decision rule is a statement of the specific conditions under which the null hypothesis is rejected and conditions under which it is not rejected. The region or area of rejection defines the location of all those values that are so large or so small that the probability of their occurrence under a true null hypothesis is rather remote.

Critical Value is the dividing point between the region where the null hypothesis is rejected and region where it is not rejected.

Step 5: Make a Decision

The fifth and final step in hypothesis testing is computing the test statistic, comparing it to the critical value, and making a decision to reject or not to reject the null hypothesis.

3.3. One - Tailed and Two - Tailed Tests of Significance

- ✓ A hypothesis could be directional or non-directional.

A *directional hypothesis* is one in which the population parameter is structured to be greater than or equal to or less than or equal to a specified value. This is known as a *one-tailed test (one-sided test)* in the parlance of statistical hypothesis.

A *non-directional hypothesis* is one in which the population parameter is structured to be equal to a specified value. This is known as a *two-tailed test (two-sided test)*.

3.4. Hypothesis Testing of Population Mean

This topic explains how to conduct a hypothesis test of a mean when the following conditions are met:

- The sampling method is simple random sampling.
- The sample is drawn from a normal or near-normal population.

The table below shows three sets of hypotheses. Each makes a statement about how the populations mean μ is related to a specified value μ_0 . (In the table, the symbol $\neq$ means "not equal to".)

Set	Null hypothesis	Alternative hypothesis	Number of tails
1	$\mu = \mu_0$	$\mu \neq \mu_0$	2
2	$\mu \geq \mu_0$	$\mu < \mu_0$	1
3	$\mu \leq \mu_0$	$\mu > \mu_0$	1

The first set of hypotheses (Set 1) is an example of a two-tailed test, since an extreme value on either side of the sampling distribution would cause a researcher to reject the null hypothesis. The other two sets of hypotheses (Sets 2 and 3) are one-tailed tests, since an extreme value on only one side of the sampling distribution would cause a researcher to reject the null hypothesis.

After specifying α we have the following regions (critical and acceptance) on the standard normal distribution corresponding to the above three hypothesis.

Summary table for decision rule

H_1	Reject H_0 if	Accept H_0 if
$\mu \neq \mu_0$	$Z_{cal} \leq -Z_{\alpha/2}$ or $Z_{cal} \geq Z_{\alpha/2}$	$Z_{cal} > -Z_{\alpha/2}$ or $Z_{cal} < Z_{\alpha/2}$
$\mu < \mu_0$	$Z_{cal} \leq -Z_{\alpha}$	$Z_{cal} > -Z_{\alpha}$
$\mu > \mu_0$	$Z_{cal} \geq Z_{\alpha}$	$Z_{cal} < Z_{\alpha}$

Where: $Z_{cal} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Examples: Left-Tail Test

Example 1: A trade commission periodically conducts studies designed to test the claims manufacturers make about their products. A level of a large can of coffee state that the mean weight is at least 3 pounds of coffee. A random sample of 36 cans of coffee is taken and the mean weight is found to be 2.92 pounds. The population standard deviation is 0.18 pounds. Do a hypothesis test at one percent level of significance.

Solution

Step 1: Structure of the hypothesis

$$H_0: \mu \geq 3$$

$$H_1: \mu < 3$$

$$n = 36, \bar{X} = 2.92, \sigma = 0.18$$

$$\sigma_{\bar{x}} = \sigma/\sqrt{n} = 0.18/\sqrt{36} = 0.03$$

Step 2: The significance level: $\alpha = 0.01$

Step 3: The test - statistic

$n = 36 > 30$, therefore, we apply z - test.

$$\alpha = 0.01$$

$$z_{0.01} = -2.33$$

$$= \frac{2.93 - 3}{0.03} = -2.67$$

Calculated z $\leq$ Critical value

$$-2.67 \leq -2.33$$

Step 4: Decision Rule

Reject H_0 if $Z_{cal} \leq -Z_{\alpha}$

Step 5: Make the Decision: Reject H_0 or H_1 is accepted!

$$Z_{cal} = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Example 2: A manager of a chain of store selling dairy products assert that the mean content the milk containers is at least 32 pounds. Perform a hypothesis test at the 1% level of significance if the mean content of a random sample of 60 containers is 31.98 pounds and the sample standard deviation is 0.1 pound.

Solution:

Step 1: $H_0: \mu \geq 32$

$H_1: \mu < 32$

$n = 60, \bar{X} = 31.98, s = 0.1$

$$s_{\bar{x}} = s / \sqrt{n} = 0.1 / \sqrt{60} = 0.0129$$

Step 2: $\alpha = 0.01$

Step 3: Test - statistic, $n \geq 30$, Apply z - test

Examples: Right-Tail Test

Example 1: A large department stock check issues its credit card. The research director is to find out if the mean monthly unpaid balance is at most 400 birr. The level of significance is 5%. A random sample of 172 unpaid balances revealed the sample mean to be 407 birr and the standard deviation of the sample to be 38 birr. Perform a test.

Step 1:

$H_0: \mu \leq 400$

$H_1: \mu > 400$

$n = 172, \bar{X} = 407, s = 38$

$$s_{\bar{x}} = s / \sqrt{n} \quad z_{0.45} = 1.64$$

$$= 38 / \sqrt{172} = 2.9$$

Step 4: Decision Rule

Reject H_0 if $z_{cal} \leq -Z_{\alpha}$

$$Z_{cal} = \frac{\bar{x} - \mu_0}{s_{\bar{x}}} = \frac{31.98 - 32}{0.0129} = -1.55$$

$$-1.55 > -2.33$$

Step 5: Pass a Decision: **Don't Reject H_0 !**

Step 2: $\alpha = 0.05$,

Step 3: Test - statistic

$n = 172 \geq 30$ Use z -test

Step 4: Decision Rule

Reject H_0 if $z_{cal} \geq Z_{\alpha}$

$$Z_{cal} = \frac{\bar{x} - \mu_0}{s_{\bar{x}}} = \frac{407 - 400}{2.9} = 2.42$$

$$2.42 \geq 1.64$$

Step 5: Decision: Reject H_0 .

Example 2: A company that specializes in products for home gardening has developed a new plant food that has been designed to increase the growing height of plant. The new plant food is tested on a sample of 12 plants of a type known to have a mean growing height of at most 18 inches. Results showed a sample mean height of 19.4 inches and a sample standard deviation of 3 inches. Assume a normal distribution and using a 10% level of significance is there any reason to believe that the new plant food increases plant height?

Solution:

Step 1: $H_0: \mu \leq 18$

$H_1: \mu > 18$

$n = 12, \bar{X} = 19.4, s = 3$

$$s_{\bar{x}} = s/\sqrt{n} = 3/\sqrt{12} = 0.866$$

Step 2: $\alpha = 10\% = 0.10$

Step 3: Test statistic $n = 12 < 30$, use t - test

Step 4: Decision Rule

$$t_{cal} = \frac{\bar{x} - \mu_0}{s_{\bar{x}}} = \frac{19.4 - 18}{0.866} = 1.62$$

$$(t_{\alpha}, v) = (t_{0.1}, n - 1)$$

Critical value = $(t_{0.1}, 11) = 1.363$

Reject H_0 if cal. $t \geq t_{\alpha, v}$

$$1.62 \geq 1.363$$

Step 5: Decision Reject H_0

Examples: Two - Tailed Test

Example 1: A manufacturer of a new cheaper type of light bulbs claims that his products just as well made and just as reliable as the higher priced competitive light bulbs. The average life of the other light bulbs is known to be 5000 hrs. In order to examine the manufacturers claim 50 of his bulbs are left on until they turn on. The average length of life in sample is 5100 hrs with a 5% level of significance. Assume that the population std. dev. is 500 hrs. Is either sufficient evidence to reject the manufacturers claim?

Solution:

Step 1: Structure of the hypothesis: $H_0: \mu = 5000\text{hrs}$ $n = 50, \alpha = 0.05, \alpha/2 = 0.025, \sigma = 500$

$H_1: \mu \neq 5000\text{hrs}$ $\sigma_{\bar{x}} = \sigma/\sqrt{n} = 500/\sqrt{50} = 70.711$

Step 2: Significance level
 $\alpha = 0.05, \alpha/2 = 0.025$

Step 4: Decision Rule

Reject H_0 if $Z_{cal} \leq -Z_{\alpha/2}$ or $Z_{cal} \geq Z_{\alpha/2}$

Step 3: Test statistic

$n = 50 > 30$ use z - test

$$Z_{0.025} = -1.96$$

$$Z_{cal} = \frac{\bar{x} - \mu_0}{\frac{\sigma_x}{\sqrt{n}}} = \frac{5100 - 5000}{\frac{70.71}{\sqrt{50}}} = 1.414$$

$$1.414 > -1.96$$

Step 5: Make a Decision:

Don't Reject H_0 .

Exercises

1. A workers' union is on strike for higher wages. The union claims that the mean salary for teachers is at most Birr 8,400 per year. The legislator does not want to reject the union's claim, however, unless the evidence is very strong against it. Assume that salaries follow a normal distribution and the population standard deviation is known to be Birr 3000. A random sample of 64 teachers is obtained, and the sample mean is Birr, 9,400. Test if the state legislator accepts the unions' claim or not at 1% significance level.
2. A fertilizer company claims that the use of its product will result in a yield of at least 35 quintals of wheat per hectare, on average, Application at the fertilizer to a randomly selected 36 sample hectares resulted in a yield of 34quintals per hectare. Assume the population standard deviation is 5 quintals and yields per hectare are normally distributed. Test the company's claim at 1% level of significance.
3. Bridge stone claims that its tires have a mean life of 35,000 miles. A random sample of 16 of these tires is tested if the sample mean is 33,000 miles. Assume that the population standard deviation is 3000 miles and the lives of tires are approximately normally distributed. Test the company's claim using a 5% level of significance.
4. A department head assumes that workers are idle for 75 minutes or less per day. A random sample of 25 workers was taken and the mean idle time was found to be 84 minutes per day with a sample standard deviation of 20 minutes. Assuming that the population is approximately normally distributed, using a 5% level of significance to test the contractor's assumption.

5. A director of a secretarial school claims that its graduates can type at least 50 words per minute on average. Suppose you want to hire some of these graduates if the director's claim is true; and you test the typing speed of 18 of the graduates and obtain a mean of 40 words per minute with a sample variance of 720. Assuming the typing speed for the graduates of the secretarial school is normally distributed, test the director's claim and decide whether to hire the graduates or not, using a 5% level of significance.
6. A magazine claims that 25% of its readers are college students. A random sample of 200 readers is taken. It is found that 42 of these readers are college students. Use a 10% level of significance and test the magazine's claim.
7. Suppose that an economist states that greater than 35% of Addis's labour force is unemployed. You don't know if the economist's estimate is too high or too low. Thus, you want to test the economist's claim using a 5% level of significance. You obtain a random sample of 400 people in the labour force, of whom 128 are unemployed. Would you reject the economist's claim?